

NLP-Based Sentiment Analysis of Movie Reviews Using TF-IDF and Logistic Regression

Mini NLP Project Report

Domain: Natural Language Processing (NLP)

Dataset: IMDb Movie Reviews Dataset

Techniques: Text Preprocessing, TF-IDF

Machine Learning Model: Logistic Regression

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Problem Statement

Movie reviews contain valuable information about viewers' opinions and experiences. However, manually analyzing a large number of reviews to determine whether they express positive or negative opinions can be time-consuming.

The objective of this project is to develop an NLP-based sentiment analysis system that automatically classifies IMDb movie reviews into two sentiment categories: **Positive** and **Negative**.

The system uses text preprocessing to clean the reviews, **TF-IDF (Term Frequency-Inverse Document Frequency)** to convert textual information into numerical features, and **Logistic Regression** as the classification algorithm. The trained model is evaluated using Accuracy, Precision, Recall, F1-Score, and a Confusion Matrix.

An interactive **Gradio-based application** is also developed to allow users to enter a movie review and receive the predicted sentiment along with a confidence score.

Objectives

The major objectives of this project are:

1. To understand the application of Natural Language Processing in sentiment analysis.
2. To preprocess and clean raw movie review text for machine learning.
3. To convert textual data into numerical features using **TF-IDF**.
4. To develop a sentiment classification model using **Logistic Regression**.
5. To classify movie reviews into **Positive** and **Negative** sentiment categories.
6. To evaluate the performance of the model using Accuracy, Precision, Recall, F1-Score, and Confusion Matrix.
7. To test the trained model on new and unseen movie reviews.
8. To develop an interactive Gradio interface for real-time sentiment prediction.

Dataset

Dataset Name

The **IMDb Movie Reviews Dataset** is used for this project. It contains movie reviews along with their corresponding sentiment labels.

Source

The dataset used in the implementation is provided as **IMDB Dataset.csv** and contains labelled movie reviews for binary sentiment classification.

Dataset Size

The dataset contains **50,000 movie reviews**.

The dataset is balanced between the two sentiment categories:

- **25,000 Positive reviews**
- **25,000 Negative reviews**

For this project, the complete dataset was loaded into a Pandas DataFrame and divided into training and testing sets using an **80:20 split**.

Therefore:

- **Training data:** 40,000 reviews
- **Testing data:** 10,000 reviews

Attributes

The main attributes used in the dataset are:

Attribute	Description
review	Textual content of the movie review
sentiment	Sentiment label of the review

An additional column called **cleaned_review** was created during preprocessing, and the sentiment was converted into a numerical **label** column.

Classes / Categories

The project performs binary sentiment classification.

Sentiment	Numerical Label
Negative	0
Positive	1

Data Preparation

The following steps were performed to prepare the dataset:

1. The dataset was loaded using Pandas.
2. The structure and sentiment distribution of the dataset were examined.
3. Movie reviews were converted to lowercase.
4. HTML tags were removed.
5. Unnecessary characters and punctuation were removed.
6. Extra spaces were removed.
7. Sentiment labels were converted into numerical values.

8. The dataset was divided into training and testing sets using an 80:20 split.
9. TF-IDF was applied to convert the cleaned reviews into numerical feature vectors.

Screenshot 1 — Dataset preview

```
# Display the first 5 rows
df.head()
```

	review	sentiment
0	One of the other reviewers has mentioned that ...	positive
1	A wonderful little production. The...	positive
2	I thought this was a wonderful way to spend ti...	positive
3	Basically there's a family where a little boy ...	negative
4	Petter Mattei's "Love in the Time of Money" is...	positive

Sample records from the IMDb Movie Reviews Dataset

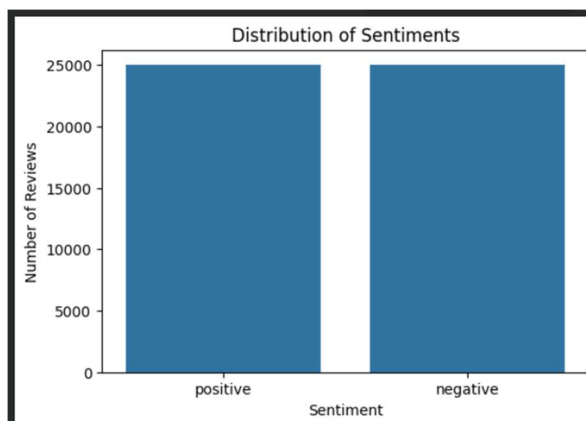
Screenshot 2 — Dataset information

```
# Display dataset information
df.info()

<class 'pandas.core.frame.DataFrame'>
RangeIndex: 50000 entries, 0 to 49999
Data columns (total 2 columns):
#   Column      Non-Null Count  Dtype
---  -
0   review      50000 non-null  object
1   sentiment   50000 non-null  object
dtypes: object(2)
memory usage: 781.4+ KB
```

Structure and attributes of the dataset

Screenshot 3 — Sentiment distribution



Distribution of sentiment classes in the dataset

Methodology

The proposed system follows a sequence of Natural Language Processing and machine learning steps to classify movie reviews according to their sentiment.

Overall Workflow

Movie Reviews → Text Preprocessing → TF-IDF Feature Extraction → Logistic Regression → Sentiment Prediction → Model Evaluation → Interactive Application

Text Data

The IMDb dataset contains movie reviews written in natural language. Each review is associated with either a Positive or Negative sentiment label.

Text Preprocessing

The raw reviews are cleaned before being given to the machine learning model. The preprocessing stage includes:

- Converting text to lowercase
- Removing HTML tags
- Removing unnecessary characters
- Removing extra spaces

This produces cleaner text for feature extraction.

TF-IDF Feature Extraction

Since machine learning algorithms cannot directly process raw text, the cleaned reviews are converted into numerical feature vectors using **TF-IDF (Term Frequency-Inverse Document Frequency)**.

TF-IDF assigns importance to words based on their occurrence in a review and their frequency across the collection of reviews.

In this project, a maximum of **5,000 features** is used to keep the feature space manageable.

Classification Model

Logistic Regression is used as the classification algorithm.

The model learns patterns between the TF-IDF features and the corresponding sentiment labels in the training data.

Sentiment Prediction

After training, the Logistic Regression model predicts the sentiment of previously unseen reviews.

The output is either:

- **Positive**
- **Negative**

Model Evaluation

The model is evaluated using:

- Accuracy
- Precision
- Recall
- F1-Score
- Confusion Matrix

These measures help determine how effectively the model classifies unseen movie reviews.

Interactive Application

A **Gradio interface** was developed to provide an interactive demonstration of the trained model.

Users can enter a movie review, and the application preprocesses the text, generates TF-IDF features, predicts the sentiment, and displays the prediction along with a confidence score.

Implementation

The project was implemented using Python in the Google Colab environment.

Programming Language

Python

Python was selected because it provides extensive libraries for Natural Language Processing, machine learning, data processing, and visualization.

Libraries Used

The following Python libraries were used:

Library	Purpose
Pandas	Data loading and manipulation
NumPy	Numerical operations
Matplotlib	Data visualization
Seaborn	Confusion matrix visualization
Regular Expressions (re)	Text preprocessing
Scikit-learn	TF-IDF, Logistic Regression and evaluation
Gradio	Interactive user interface

NLP Techniques

The major NLP techniques used in the project are:

- Text preprocessing

- Lowercase conversion
- HTML tag removal
- Unnecessary character removal
- TF-IDF feature extraction

Machine Learning Algorithm

Logistic Regression was used for binary sentiment classification.

The model was trained using the TF-IDF representations of the training reviews.

Important Implementation Details

The dataset was divided into training and testing sets using an **80:20 ratio**.

The TfidfVectorizer was configured with a maximum of **5,000 features** and English stop-word removal.

The Logistic Regression model was trained with max_iter=1000.

After training, predictions were generated for the test data and evaluated using standard classification metrics.

The trained model was also integrated with Gradio to create an interactive sentiment analysis application.

Results

The trained Logistic Regression model was evaluated using the test dataset. The model achieved good performance across all the evaluation metrics.

Performance Metrics

Evaluation Metric	Score
Accuracy	88.66%
Precision	87.86%
Recall	89.72%
F1-Score	88.78%

The model achieved an overall **accuracy of 88.66%**, indicating that the majority of the unseen movie reviews were correctly classified.

The Precision score of **87.86%** indicates that a high proportion of reviews predicted as positive were actually positive.

The Recall score of **89.72%** indicates that the model successfully identified a high proportion of the actual positive reviews.

The F1-Score of **88.78%** shows a good balance between Precision and Recall.

Confusion Matrix

The confusion matrix provides a visual representation of the correct and incorrect classifications made by the model. It shows the relationship between the actual sentiment labels and the predicted sentiment labels.

Testing with New Reviews

The trained model was also tested using new movie reviews that were not part of the training dataset.

The system successfully generated sentiment predictions for the sample reviews and displayed the predicted sentiment along with a confidence score.

Interactive Application

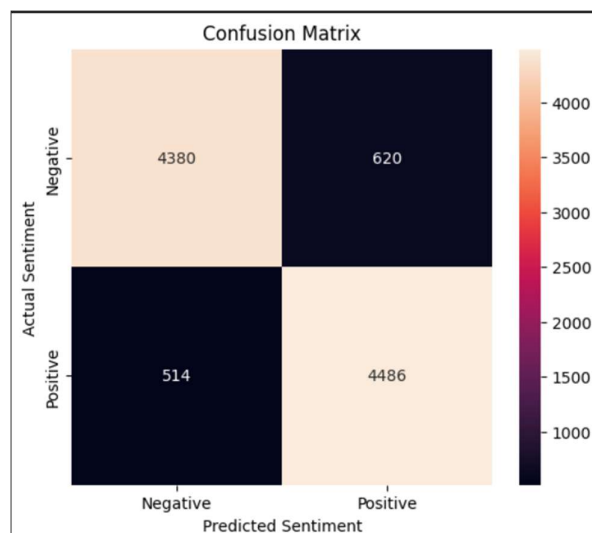
A Gradio-based interface was developed to demonstrate the practical use of the trained model. Users can enter a movie review and obtain the predicted sentiment and confidence score through the interface.

Classification report of the sentiment classification model:

```
# Display detailed classification report
print(classification_report(
    y_test,
    y_pred,
    target_names=["Negative", "Positive"]
))
```

	precision	recall	f1-score	support
Negative	0.89	0.88	0.89	5000
Positive	0.88	0.90	0.89	5000
accuracy			0.89	10000
macro avg	0.89	0.89	0.89	10000
weighted avg	0.89	0.89	0.89	10000

Confusion matrix of the Logistic Regression model:



Performance comparison of evaluation metrics:

	Metric	Score	
0	Accuracy	0.8866	
1	Precision	0.8786	
2	Recall	0.8972	
3	F1-Score	0.8878	

Interactive IMDb Sentiment Analysis application:

 **IMDb Movie Review Sentiment Analyzer**

Enter a movie review to predict whether it is Positive or Negative.

Movie Review

The movie was really good

Clear

Submit

Prediction

Sentiment: Positive
Confidence: 82.36%

Flag

Examples

This movie was absolutely fantastic. The acting was brilliant...

The movie was boring and disappointing. I hated it.

An amazing story with excellent performances.

Limitations

Although the developed system provides good sentiment classification performance, it has some limitations:

1. Limited Domain:
The model is trained specifically on movie reviews. Its performance may vary when applied to other types of text such as product reviews, social media posts, or news articles.

2. **Limited Context Understanding:**
TF-IDF represents words based on their statistical importance but does not fully understand the deeper context or meaning of a sentence.
3. **Difficulty with Sarcasm and Irony:**
The model may incorrectly classify reviews containing sarcasm, irony, or indirect expressions.
4. **Binary Classification:**
The system supports only Positive and Negative sentiment and does not identify neutral or more detailed emotional categories.
5. **Unseen Expressions:**
Reviews containing unfamiliar words, slang, or unusual expressions may result in incorrect predictions.
6. **Preprocessing Limitations:**
Removing punctuation and other characters may sometimes remove information that could contribute to sentiment interpretation.

Future Scope

The project can be further improved and extended in several ways:

1. **Larger and More Diverse Datasets:**
The system can be trained using larger datasets containing reviews from different domains.
2. **Advanced NLP Models:**
Word embeddings and transformer-based models such as BERT can be explored to achieve better contextual understanding.
3. **Multi-Class Sentiment Analysis:**
The system can be extended to classify reviews as Positive, Negative, or Neutral.
4. **Emotion Detection:**
Future versions can identify emotions such as happiness, anger, sadness, fear, or excitement.
5. **Improved Handling of Sarcasm:**
Specialized techniques can be introduced to better handle sarcasm, irony, negation, and mixed sentiments.
6. **Multilingual Support:**
The system can be extended to support reviews written in multiple languages.
7. **Web or Mobile Deployment:**
The model can be deployed as a web or mobile application for real-time sentiment analysis.
8. **Integration with Other Systems:**
The sentiment analysis system can be integrated with movie recommendation or review analysis platforms.

Conclusion

This project successfully developed an NLP-based sentiment analysis system for IMDb movie reviews using text preprocessing, TF-IDF feature extraction, and Logistic Regression.

The system automatically classifies movie reviews into Positive and Negative sentiment categories. The trained model achieved an Accuracy of 88.66%, Precision of 87.86%, Recall of 89.72%, and F1-Score of 88.78% on the test dataset.

The project also demonstrated the practical application of the trained model through an interactive Gradio interface, where users can enter movie reviews and receive sentiment predictions with confidence scores.

Through this project, the importance of text preprocessing, feature extraction, machine learning classification, and model evaluation in Natural Language Processing was understood. The project also provided practical experience in developing an end-to-end NLP application, from raw text processing to interactive prediction.